

PRECAL

Pre-Calculus

Grade 12, Semester 1 study outline

A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.

01. The Unit Circle and Evaluating Trigonometric Functions

No outline could be read from this handout.

02. Coterminal Angles and Conversion- Degrees - Radian Measure

1. Find one positive angle and one negative angle that are coterminal with -45°
2. Find one positive angle and one negative angle that are coterminal with 395°
3. Find one positive angle and one negative angle that are coterminal with $\frac{1}{6}$ radians
4. Find two positive angles that are coterminal with $\frac{11}{2}$ radians

Note: Since $\frac{11}{2}$ is more than one complete revolution, 2 was subtracted from the initial angle yielding a

1. Convert 120° to radians
2. Convert 110° to radians

The angle 110° is between the special angles 90° and 120° . The angle $\frac{11}{18}$ is between the same special angles:

3. Convert $-\frac{1}{12}$ radians to degrees
4. Convert $-\frac{1}{9}$ radians to degrees

The angle $-\frac{1}{9}$ is between the special angles 0° and $-\frac{1}{6}$. The angle -20° is between the same special angles: 0°

1. Find one positive angle and one negative angle that are coterminal with 130°
2. Find one positive angle and one negative angle that are coterminal with -60°
3. Find one positive angle and one negative angle that are coterminal with 525°
4. Find one positive angle and one negative angle that are coterminal with 645°
5. Find one positive angle and one negative angle that are coterminal with $\frac{13}{18}$
6. Find one positive angle and one negative angle that are coterminal with $-\frac{2}{3}$

Key terms

Positive angles - generated by counterclockwise rotation, and negative angles by clockwise rotation, as shown in

Angles - labeled with Greek letters like alpha, beta, and theta, as well as uppercase letters

Coterminal Angles - angles who share the same initial side and terminal sides. Finding coterminal angles is as

03. Sum and Difference Identity(Sine Cosine)

1. Write the formula for sine
2. Substitute the given angles into the formula
3. Find the values of the special angles
4. Substitute the values into the equation
5. Simplify

1. $\sin(45^\circ - 30^\circ)$

$$\sin(45^\circ - 30^\circ) = \sin(45^\circ) \cos(30^\circ) - \cos(45^\circ) \sin(30^\circ)$$

2. $\sin(45^\circ + 30^\circ)$

$$\sin(45^\circ + 30^\circ) = \sin(45^\circ) \cos(30^\circ) + \cos(45^\circ) \sin(30^\circ)$$

3. $\sin(135^\circ - 120^\circ)$

$$= \sin(135^\circ) \cos(120^\circ) - \cos(135^\circ) \sin(120^\circ)$$

1. Write the formula for sine

2. Substitute the given angles into the formula
3. Find the values of the special angles
4. Substitute the values into the equation
5. Simplify
2. Find the exact value of $\cos(75^\circ)$

STEP 3, 4 & 5: Find and substitute the values of the trigonometric expressions then simplify.

04. Sum and Difference Identity(Tangent)

1. Write the formula for the tangent
2. Substitute the given angles into the formula
3. Simplify
1. Find the exact value of

Let's first write the sum formula for tangent and then substitute the given angles into the formula.

2. Find the exact value of $\tan 75^\circ$
3. Find the exact value of $\tan 105^\circ$

Finding Multiple Sums and Differences of Angles

1. $\tan 75^\circ$

Note that we can express as . Hence, using the formula

2. $\tan 165^\circ$
4. Given

From the given, both and are in the third quadrant.

05. Double Angle Identity(Cosine)

1. Given
2. Given
3. Given
4. Given
5. Given

Similar to Example No. 2, we can solve this item in several ways. We know from the formula that

06. Double Angle Identity(Sine and Tangent)

1. Given
2. Given
3. Given
4. Given

From the previous lesson, using the Pythagorean identity, we can get . Now, using the formula

Again, using the Pythagorean identity from the previous discussion, we can get . Then, we

Note that from the given, we can calculate using pythagorean identity. Then we can solve

Note that from the given, we can calculate using Pythagorean identity. Then we can solve

07. Half Angle Identity(Sine and Cosine)

1. Find the exact value of $\sin(15^\circ)$ using a half-angle formula
2. Find the exact value for $\sin 105^\circ$ using the half-angle identity

Since $15^\circ = 30^\circ/2$,we use the half-angle formula for sine

Note that 105° is in the second quadrant, and sine functions in the second quadrant are positive. Also, 210° is in

1. Find the exact value of $\cos(15^\circ)$ using a half-angle formula

Since the angle 15° is in the first quadrant, where the cosine is positive, the value is

2. Find the exact value for $\cos 165^\circ$ using the half-angle identity

Note that 165° is in the second quadrant, and cosine functions in the second quadrant are negative. Also, 330° is

3. Use a Half Angle Identity to find the exact value of $\cos(18^\circ)$

Since 18° is in the first quadrant, we will use the positive square root. We can also rewrite the expression under

1. Given

If $\theta = 18^\circ$, then $2\theta = 36^\circ$. Since 36° is in the first quadrant, $\sin(36^\circ)$ is also in the first quadrant, where sine is

2. Given

3. Given

We can use the half-angle identity directly to find the $\cos(18^\circ)$. Since 36° is in Quadrant IV, then $\sin(36^\circ)$ will be in the

4. Given

From the previous lessons, use Pythagorean identity and the given to find $\cos(18^\circ)$. We take the

08. Half Angle Identity(Tangent)

1. Given

First, we need to determine the sine and cosine of θ . Use the concepts from previous lessons to obtain the

2. Given

First, we need to determine the sine of θ . Use the concepts from previous lessons to obtain $\sin(\theta)$.

3. Given

Similar to Example No. 2, we need to determine the sine of θ . Using the concepts from previous lessons to

4. Given

Repeat the process from the previous examples.

09. Conic Section Classification

1. Given

Recall: Note that the general form of conic sections is denoted by

2. Given

3. Given

4. Given

5. Given

We can observe that both x and y are squared, with the same coefficients and sign. Thus, the conic section

6. Given

The conic section is a PARABOLA, since only x is squared.

7. Given

Since x and y are both squared, and their coefficients are both positive but with different values, the conic

8. Given

Notice that x and y are both squared, and one of them has a negative coefficient, and the other one has a

Key terms

PARABOLA - formed when the plane is parallel to the surface of the cone, resulting in a U-shaped curve that lies

Here's - some more important pointers when we're given the general form of a conic section equation

Certain characteristics - unique to each type of conic and hint at which of the conic sections you're graphing. To

Since x and y - both squared, and their coefficients are both positive but with different values, the conic

10. Circle(Part 1)

1. Find the general equation of the circle

2. State the radius and center of the circle with equation

3. Find the general equation of a circle with center and radius are

4. Find the general equation of a circle

First, we compute the radius using the given and the distance formula:

5. Find the general equation of a circle

First, we compute the radius using the given and the distance formula:

Key terms

CIRCLE - collection of points equidistant from a fixed point. The fixed point is called the center. The distance

11. Circle(Part 2)

1. Given
 2. Given
 3. Given
 4. Given
-

12. Parabola(Part 1)

1. Given
2. Given
3. Given
4. Given
5. Given
6. Given
7. Given
8. Given

Topic: Parabola - General Form to Standard Form and Discussion of Parabola's Equation

Key terms

Since a - undirected distance between the vertex and the focus, it is equal to $3/4$

Since the vertex - midway between the focus and directrix, its coordinate is at the origin $(0, 0)$. Substituting in the

13. Parabola(Part 2)

1. Given
2. Given
3. Given

Rewrite in standard form: . This is in the form where so .

4. Find the general equation of the parabola that satisfies the given condition

Since the x-coordinates of both the vertex and focus are the same, the orientation of the parabola is

5. Find the general equation of the parabola that satisfies the given condition

Solution: Since the directrix is vertical, the parabola opens horizontally. The parabola opens to the right

6. Find the general equation of the parabola which satisfies the given condition

First, determine the vertex . Since the directrix is horizontal, i.e. , the parabola's axis of

14. Ellipse(Part 1)

1. Given
2. Given
3. Given
4. Given
5. Given

Topic: Ellipse - General Form and Standard Form Equation

15. Ellipse(Part 2)

1. Given
2. Given
3. Given
4. Given

16. Hyperbola(Part 1)

1. Given
2. Given
3. Given
4. Given
5. Given

Topic: Hyperbola - General Form and Standard Form Equation

17. Hyperbola (Part 2)

1. Given
2. Given
3. Given
4. Given

18. Systems of Nonlinear Equations

1. Solve the linear equation for one of the variables
2. Substitute the expression obtained in step one into the parabola equation
3. Solve for the remaining variable
4. Check your solutions in both equations
1. Solve the linear equation for one of the variables
2. Substitute the expression obtained in step one into the equation for the circle
3. Solve for the remaining variable
4. Check your solutions in both equations

Find the intersection of the given circle and the given line.

4. Solve the following system $y = x^2 + 2x + 1$ and $y = 2x^2 + 3x + 2$, and sketch the graph

Note that there is a negative number inside the square root (-3) and it can't be graphed. The graphs of two